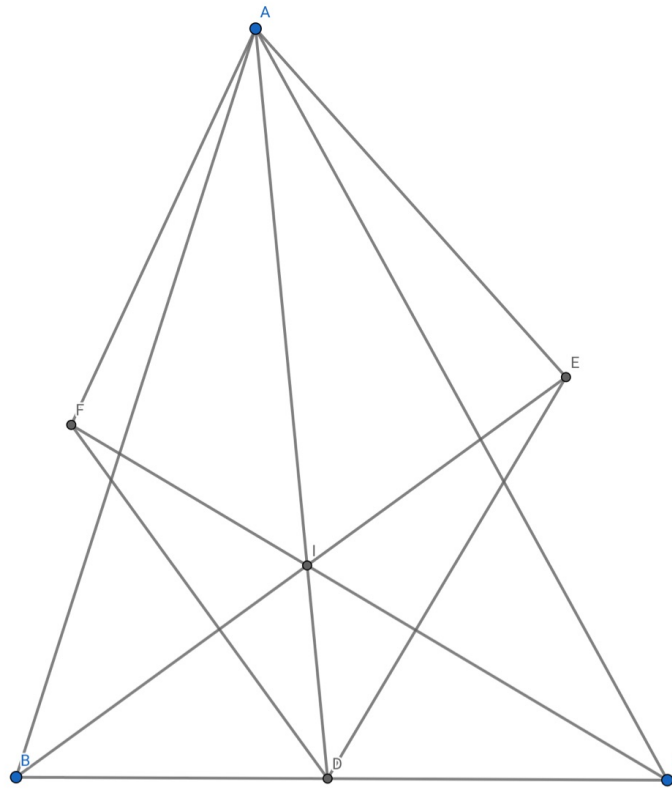


4.3.27

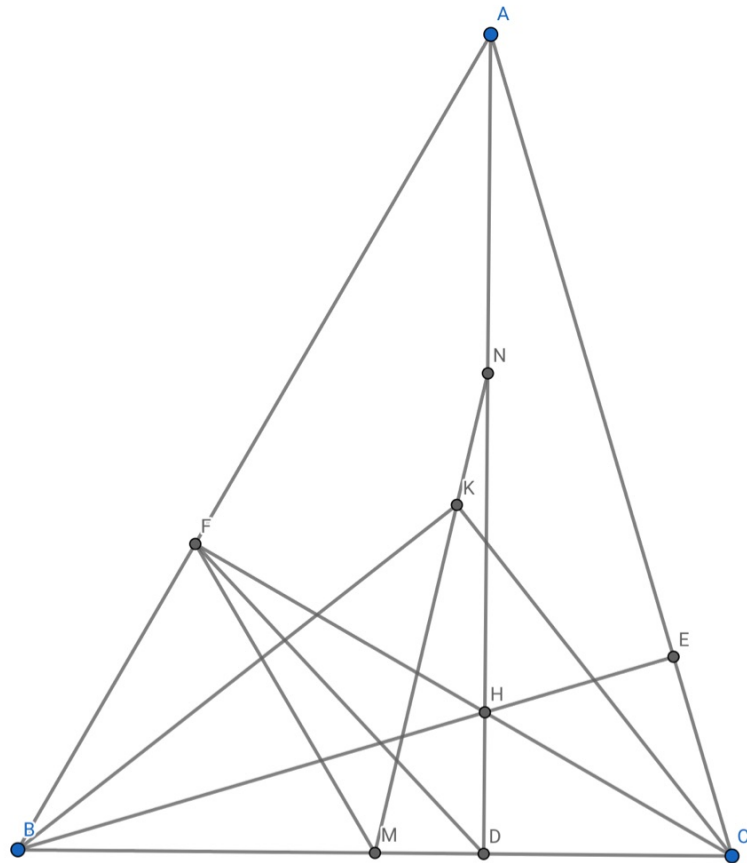


The angle bisector of $\angle ABD$ and perpendicular bisector of AD meet at O_{ABD} , so A, B, D, E are concyclic. Similarly, A, C, D, F are concyclic.

$$\begin{aligned} \therefore \angle AEI + \angle AFI &= \angle AEB + \angle AFC \\ &= \angle ADB + \angle ADC \\ &= 180^\circ. \end{aligned}$$

So A, E, I, F are concyclic.

4.3.32



Let $AD \perp BC$ at D .

$$\begin{aligned}
 \angle BKC &= \angle BHC - \angle HBK - \angle HCK \\
 &= 180^\circ - A - \frac{\angle ABE}{2} - \frac{\angle ACF}{2} \\
 &= 180^\circ - A - \frac{90^\circ - A}{2} - \frac{90^\circ - A}{2} \\
 &= 90^\circ \\
 &= \angle BFC
 \end{aligned}$$

So B, F, K, C are concyclic with centre M .

Since $\angle BDH = \angle BFH = 90^\circ$, B, D, H, F are concyclic.

By nine point circle, D, M, F, N are concyclic.

$$\therefore \angle FMK = 2\angle FBK$$

$$= \angle FBH$$

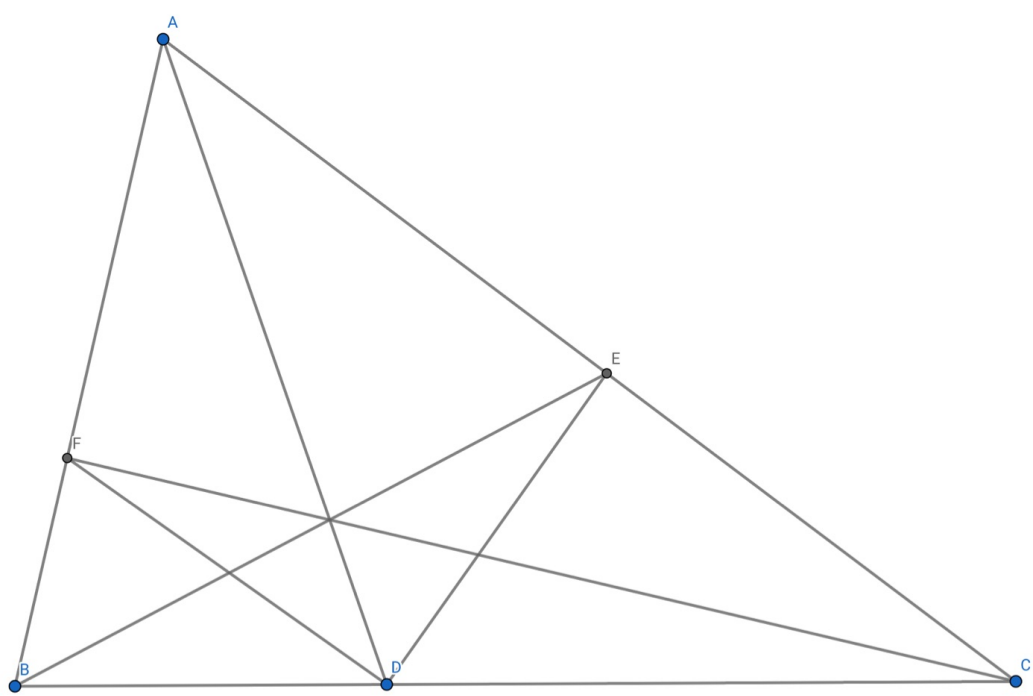
$$= \angle FDH$$

$$= \angle FDN$$

$$= \angle FMN$$

So M, K, N are collinear.

4.3.33



By angle bisector theorem,

$$\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = \frac{BD}{DC} \cdot \frac{CD}{DA} \cdot \frac{AD}{DB} = 1$$

By Ceva's theorem, AD, BE, CF are concurrent.